Benjamin Yu

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NLP researcher working with the UCLA NLP Group, Q-Lab, and SPAR. My research interests are in how intelligent behavior, such as self-awareness and introspection, arise during training, with the aim to produce a theory which predicts these behaviors. I pursue this goal through several projects, probing internal representations to establish what a model encodes but does not report (SPAR), and designing training objectives intended to produce these behaviors directly (UCLA NLP Group, Q-Lab)

Education

- **University of California, San Diego** — **La Jolla, CA**
  M.S. in Computer Science — *2026–2028*

- **University of California, Los Angeles** — **Los Angeles, CA**
  B.S. in Data Theory — *2022–2026*
  Magna Cum Laude, GPA 3.97/4.00.

Publications

- **Benjamin Yu**, Vincenzo Lordi, Daniel Schwalbe-Koda. Maximizing Efficiency of Dataset Compression for Machine Learning Potentials With Information Theory. *Journal of Chemical Physics* (accepted). arXiv:2511.10561
- **Benjamin Yu**, Jackie Liu, Justin Cui. Smart-GRPO: Smartly Sampling Noise for Efficient RL of Flow-Matching Models. *AAAI 2026 AIR-FM Workshop*. arXiv:2510.02654

Research Experience

- **SPAR (Supervised Program for Alignment Research)** — **Remote**
  Research Fellow, PI: Christopher Ackerman (MATS) — *2026–Present*
  - **Self-awareness in LLMs** (lead): studying self-awareness in language models, and building empirical measures and experiments to measure self-awareness (persistent goals and strategic behavior, identity consistency across contexts), and exploring mechanisms causing self-awareness behavior, such as post-training techniques.

- **UCLA NLP Group** — **Los Angeles, CA**
  Researcher, PI: Nanyun Peng — *2024–Present*
  - **Metacognitive LLM calibration** (lead): training large-language models to be better calibrated by using its own internal confidence, building on RL with calibration rewards (RLCR) by incorporating an internal–verbal energy alignment term
  - Designed and implemented RLICA (reinforcement learning with internal confidence alignment) on a customized veRL/GRPO/LoRA training pipeline. Trained Qwen3.5-4B-Base with this method and showed substantially better calibration than outcome-only RL baselines (eight-benchmark average ECE 0.174 vs. 0.229/0.382; AbstentionBench recall 81.5% vs. 65.8%/27.0%)
  - Ran mechanistic-interpretability experiments, including linear and sparse-autoencoder probing of the confidence signal across layers and causal steering tests

- **Q-Lab, UC San Diego** — **La Jolla, CA**
  Researcher, PI: Prof. Lianhui Qin — *2026–Present*
  - **Discrete chain-of-thought** (co-lead): improving discrete latent reasoning in large language models through synthetic data generation and interleaving reasoning for better autoregressive inductive bias
  - Designed and built a multi-stage synthetic data pipeline that converts verified rStar-Coder solutions into interleaved latent-reasoning/natural-language training traces, testing an execution-guided revision approach and a plan-then-branch approach before adopting a distillation-based pipeline

UCLA Computational Machine Learning Group **Los Angeles, CA**
Undergraduate Researcher, PI: Prof. Justin Cui *2025*

- **Smart-GRPO** (lead): Led an early-stage project on noise optimization for RL of flow-matching generative models.
- Implemented the noise-search step as an iterative cross-entropy-method-style refinement. Trained SD3.5-Medium using ImageReward and Aesthetic Score as reward signals, improving ImageReward from 0.824 (Flow-GRPO) to 0.858 and Aesthetic Score from 6.111 to 6.238, with faster reward convergence over training.

UCLA Digital Synthesis Lab **Los Angeles, CA**
Undergraduate Researcher, PI: Prof. Daniel Schwalbe-Koda *2024–2025*

- **Efficient Compression of Atomistic Datasets** (lead): Information-theoretic compression of atomistic datasets for ML interatomic potentials, framed as a minimum set-cover problem; compressed sets preserve diversity and reduce error on out-of-distribution data (open-source QUESTS package). First-author, J. Chem. Phys. (accepted).
- Formulated atomistic dataset compression as minimum set cover over atom-centered environments, and solved the NP-hard cover greedily.
- Benchmarked against random, k-means, and farthest-point sampling on GAP-20, TM23, and 64 ColabFit datasets, reaching near-lossless (99%) environment overlap at 10% compression for redundant elements, up to 30 points higher overlap than the next-best baseline, and better-preserved long-tail force-error distributions. Codebase released as the open-source QUESTS package.

Experience

Johns Hopkins Applied Physics Laboratory **Laurel, MD**
Machine Learning Intern *2025*

- Built retrieval-augmented generation (RAG) over 300+ page documents using LangChain and transformers.
- Fine-tuned encoder models (CrossEncoders, SBERT) with LoRA for failure-log classification.
- Developed MLP, CNN, and diffusion models to predict gravitational fields using PyTorch and diffusers.

U.S. Naval Research Laboratory **Washington, DC**
Software Engineer Intern *2024*

- Devised a keypoint-detection model locating a matrix of targets within 5% error for 95% of data.
- Implemented complex-valued neural networks classifying raw satellite data, reaching 97% validation accuracy.

Grants

Rapid Grant – BlueDot Impact, $3,500 (2026). Funding for independent research on safe and beneficial AI.

Leadership

Workshop Officer & AI Scholarship Foundation Co-Director – ACM AI at UCLA (2025–2026).